

Learning Objectives

After studying this chapter, the student will be able to:

- ➡ Understand the principle of laser physics
- ➡ Know about the different types of Laser and their selected applications
- ➡ Recall the principle of hologram formation and reconstruction
- ➡ Understand few important industrial and medical applications of laser
- ➡ Learn how to use the holographic techniques for different purposes.

6.1 INTRODUCTION

LASER is an acronym for Light Amplification by Stimulated Emission of Radiation. In laser, the intensity of light is amplified by a process called stimulated emission.

Lasers are optical phenomena which find major application in various fields such as medicine, engineering, fiber optic communication, industries, etc. Lasers are more powerful radiation than ordinary light radiation.

6.2 CHARACTERISTICS OF LASER

The following characteristics, distinguishes a laser beam from an ordinary light. They are:

1. Coherence
2. High intensity
3. High directionality
4. High monochromaticity

Laser light is highly powerful and is capable of propagating over long distances and is not easily absorbed by water.

6.2.1 Coherence

The wave trains which are identical in phase and direction are called coherent waves.

Since all the constituent photons of laser beam possess the same energy, momentum and propagate in same direction, the laser beam is said to be highly coherent.

6.2.2 High Intensity

Due to the coherent nature of laser, it has the ability to focus over a small area of 10^{-6} cm^2 , i.e., extremely high concentration of its energy over a small area.

6.2.3 High Directionality

An ordinary light source emits light in all possible directions. But, since laser travels as a parallel beam it can travel over a long distance without spreading.

The angular spread of a laser beam is 1 mm/meter. This reveals the directionality of the laser beam.

6.2.4 High Monochromaticity

The light from a normal monochromatic source spreads over a range of wavelength of the order 100 nm. But, the spread is of the order of 1 nm for laser. Hence, laser is highly monochromatic, i.e., it can emit light of single wavelength.

6.3 EINSTEIN'S THEORY

Einstein explained the action of laser beam based on quantum theory of light. Production of laser light is a particular consequence of interaction of radiation with matter. Radiation interacts with matter under appropriate conditions and may lead to the transition of an atom or a molecule from one energy state to another. If the transition is from a higher state to a lower state, the system gives a part of its energy. But, if the transition is in the reverse direction, then it absorbs the incident energy.

There are three possible ways through which interaction of radiation and matter can take place. Among the three types, one is absorption that is also known as induced absorption and the other two are emissions.

The emission of radiation can occur in two ways as suggested by Einstein. They are spontaneous emission and stimulated emission.

The interpretation of the interaction is done on the basis of ideas related to energy levels of the concerned system for which light is to be obtained.

All the three processes are described by considering an atom having only two energy levels E_1 and E_2 .

(a) Induced Absorption (Absorption)

Let the atom be initially in the lower state E_1 . If a photon of energy $h\nu$ is incident on the atom in

the lower state, the atom absorbs the incident photon and gets excited to the higher energy state E_2 . This process is called induced absorption as shown in Figure 6.1.

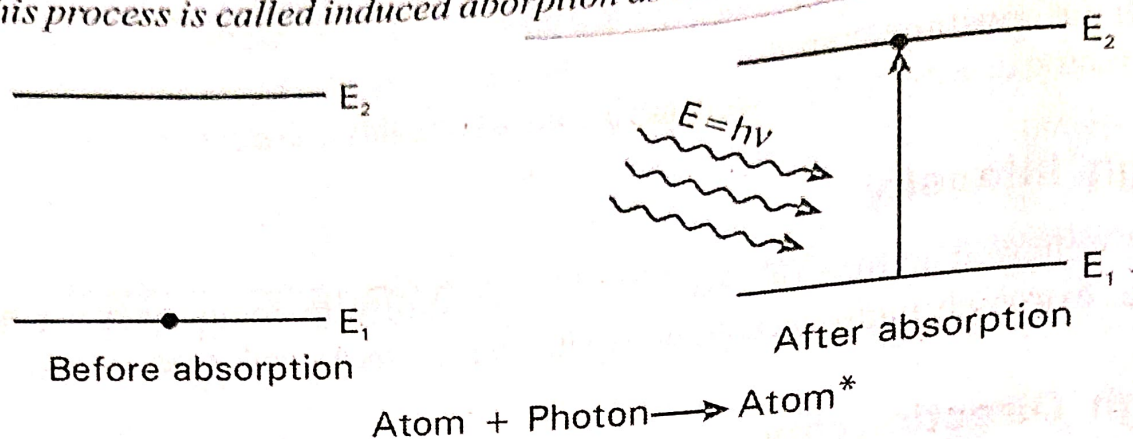


Figure 6.1: Induced absorption

→ The rate of absorption R_{12} is proportional to the population of the lower energy level N_1 and to the density of incident radiation ρ . Hence,

$$R_{12} \propto N_1 \rho \quad \dots(1)$$

or

$$R_{12} = B_{12} N_1 \rho$$

→ where B_{12} is the proportionality constant known as the probability of absorption of radiation per unit time.

(b) Spontaneous Emission

It is a process in which there is an emission of a photon whenever an atom transits from a higher energy state to a lower energy state without the aid of any external agency.

For this process to take place, the atom has to be in the excited state. Since, the higher energy level is an unstable one, the excited atom in the higher energy level E_2 spontaneously returns to the lower energy level E_1 with the emission of a photon of energy $h\nu = E_2 - E_1$ as shown in Figure 6.2.

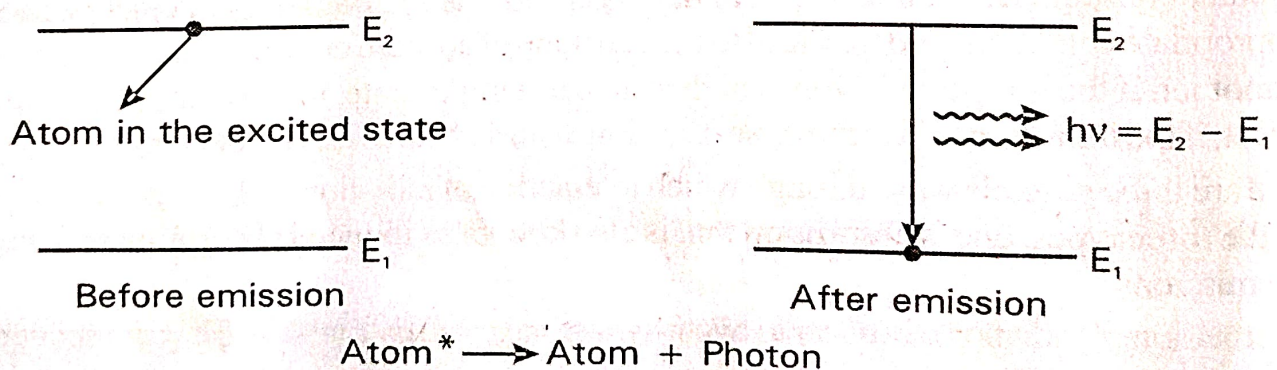


Figure 6.2: Spontaneous emission

The rate of spontaneous emission of radiation $R_{21}(\text{sp})$ is proportional to the population N_2 at the higher energy level E_2 .

i.e.,

$$R_{21}(\text{sp}) \propto N_2$$

∴

$$R_{21}(\text{sp}) = A_{21} N_2 \quad \dots(2)$$

where A_{21} is the proportionality constant known as the probability of spontaneous emission per unit time.

(c) Stimulated Emission

It is a process in which there is an emission of a photon whenever an atom transits from a higher energy state to a lower energy under the influence of an external agency, i.e., an inducing photon.

For this process also, the atom should be already in the excited state. Let a photon having an energy $h\nu = E_2 - E_1$ interact with an atom in the excited state. Under such interaction, the incident photon stimulates the excited atom in the level E_2 to transit to the lower energy level E_1 , resulting in the emission of a photon of energy $h\nu = E_2 - E_1$ as shown in Figure 6.3.

Both the inducing photon and the emitted photon will have the same phase, energy and direction of movement. This kind of emission is responsible for laser action, i.e., the stimulated emission of radiation is the principle used in laser action.

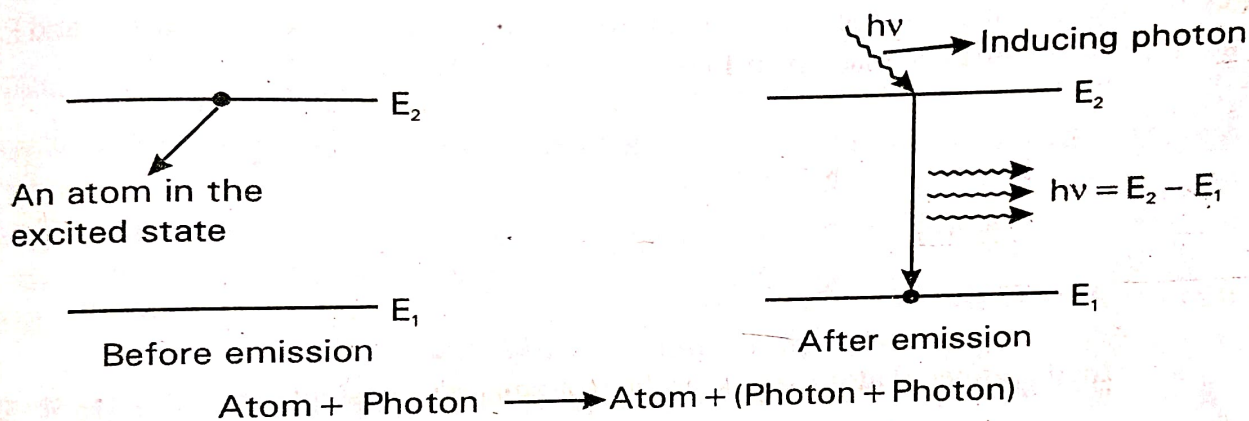


Figure 6.3: Stimulated emission

The rate of stimulated emission of radiation $R_{21}(\text{st})$ is proportional to the population N_2 at the higher energy level E_2 and to the density ρ of the inducing photon.

$$\text{i.e.,} \quad R_{21}(\text{st}) \propto N_2 \rho$$

$$\therefore R_{21}(\text{st}) = B_{21} N_2 \rho \quad \dots(3)$$

where B_{21} is the proportionality constant known as the probability of stimulated emission of radiation per unit time.

The coefficients B_{12} , A_{21} and B_{21} in the equations (1), (2) and (3) are called the Einstein's coefficients.

6.3.3 Difference between Stimulated and Spontaneous Emission

S. No.	Stimulated Emission	Spontaneous Emission
1.	Emission of a light photon takes place through an inducement i.e., by an external photon.	Emission of a light photon takes place immediately without any inducement.
2.	It is not a random process.	It is a random process.
3.	The photons get multiplied through chain reaction.	The photons do not get multiplied through chain reaction.
4.	It is a controllable process.	It is an uncontrollable process.
5.	More intense.	Less intense.
6.	Monochromatic radiation.	Polychromatic radiation.

6.4 BASIC CONCEPTS IN LASER PHYSICS

6.4.1 Population Inversion

Population inversion is a state of achieving more number of atoms in the excited state compared to the ground state.

i.e.,
$$N_2 > N_1$$

If this condition is satisfied, then there is more chance for stimulated emission to take place. Hence, population inversion is an essential condition for producing laser.

Population inversion can be achieved by a process called pumping.

6.4.2 Pumping

Pumping is the mechanism of exciting atoms from the lower energy state to a higher energy state by supplying energy from an external source.

The most commonly used pumping mechanisms are described below.

Optical Pumping

In this type of pumping atoms are excited (i.e., population inversion is achieved) by means of an external optical source.

This type of pumping technique is adopted in solid state lasers such as ruby laser and Nd: YAG laser.

Electrical Pumping (Direct Electron Excitation)

In this type of pumping the electrons are accelerated to a high velocity by a strong electric field. These moving electrons collide with the neutral gas atoms and ionise the medium. Thus, due to ionisation they get raised to a higher energy level.

This technique of pumping is adopted in gas lasers such as CO₂ laser.

Direct Conversion

In this type of pumping, a direct conversion of electric energy into light takes place. This technique of pumping is adopted in semiconductor laser.

In addition to the above three, the other types of pumping are inelastic collision between atoms and chemical methods which are, respectively, adopted in He-Ne gas laser and in dye and chemical lasers.

Lasing

The process which leads to emission of stimulated photons after establishing the population inversion is referred to as lasing.

Life Time

The limited time for which a particle or an atom remains in the excited is known as life time. It is about a nano second.

Metastable State

Metastable states are the energy levels in an atomic system where the life time of atoms is very large (of the order 10^{-3} to 10^{-2} second). This property helps in achieving the population inversion.

Active Medium

A medium in which population inversion is achieved for laser action is called active medium. The medium can be solid, liquid, gas and plasma.

Based on the active medium and method of pumping, the lasers are classified into:

- (1) Solid state lasers
- (2) Liquid lasers
- (3) Gaseous lasers
- (4) Dye lasers
- (5) Semiconductor laser

Optical Resonator

It is a pair of reflecting surfaces (mirrors) of which one is a perfect reflector and the other is a partial reflector.

It is used for amplification of photons thereby producing an intense and highly coherent output.

Principle of Laser

Laser is based on the principle of stimulated emission of radiation with light amplification. For stimulated emission of radiation to take place, the population of atoms in higher energy level should be greater than the lower energy level, i.e., $N_2 > N_1$. This can be achieved by pumping. Light amplification is achieved by photon multiplication within an optical resonator cavity.

Initially, the state of population inversion has to be achieved in the active medium which is within a resonator cavity. Then, a spontaneously emitted photon by one of the excited atom stimulates another atom it encounters in its path to release a second photon. Thus, these two photons which

Basic Laser System Components

- (1) The active medium
- (2) The pumping source
- (3) The optical resonator

6.5 Nd: YAG LASER

Active Medium

Resonator Cavity

The end faces of the Nd: YAG rod are ground polished and silvered to act as the optical resonator mirrors, or the optical cavity can be formed by using two external reflecting mirrors M_1 and M_2 .

Optical Pumping

A xenon flash lamp or a krypton flash lamp is used as a pumping source.

Construction

The schematic diagram of a Nd:YAG laser is shown in Figure 6.4.

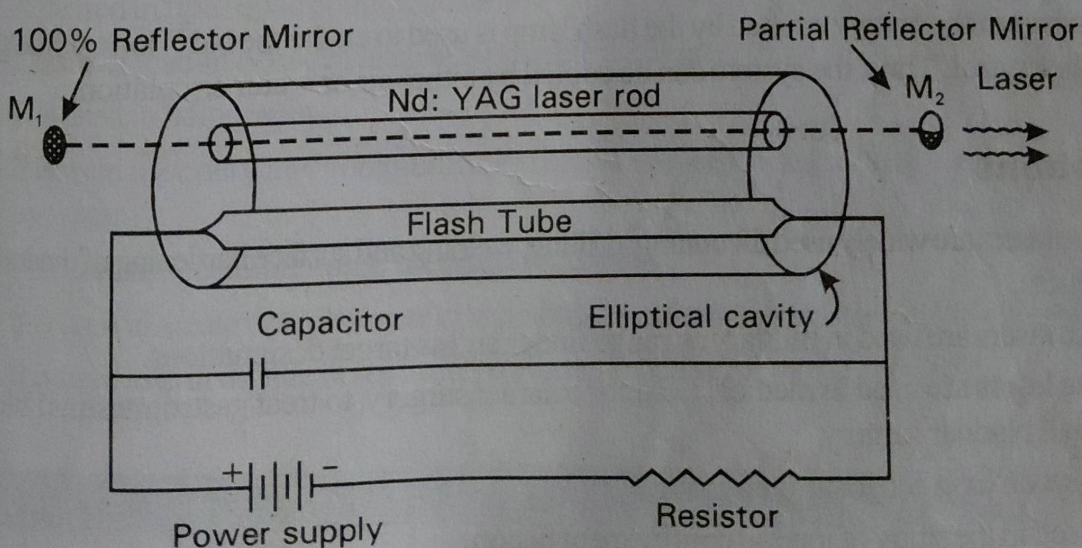


Figure 6.4: Nd: YAG laser

6.6 APPLICATIONS OF LASER

Some of the applications of laser in various fields are listed below.

(a) In Industry

In industries, lasers are applied to a larger extent for the following processes:

- (1) For welding and melting.
- (2) For cutting and drilling holes.
- (3) To test the quality of the materials.
- (4) For the heat treatment of metallic and non-metallic materials.

(b) In Medicine

- (1) Used for the treatment of detached retinas.
- (2) Used in performing micro and bloodless surgery.
- (3) Used for the treatment of human and animal cancers and skin tumours.

(c) Military applications

- (1) The laser beam can serve as a war weapon, i.e., A powerful laser beam can be used to destroy in a few seconds, the big size objects like aeroplanes, missiles etc., by pointing the laser beam on to them. For this reason, it can be even called as death ray.
- (2) The laser beam can be used to determine precisely the distance, velocity and direction as well as the size and form of distant objects by means of the reflected signal. It is known as LIDAR.

(d) Science and engineering applications

- (1) It is used in fiber optic communication.
- (2) Communication between planets is possible with laser.
- (3) It is used in holography.
- (4) It is used in underwater communication between submarines, as they are not easily absorbed by water.
- (5) It is used to accelerate some chemical reactions.
- (6) It is used to create new chemical compounds by destroying atomic bonds between molecules.
- (7) It is used to drill minute holes in cell walls without damaging the cell itself.

6.7 LASERS IN MATERIAL PROCESSING

Laser has widespread nature of material processing. In this section few important examples of laser in material processing are discussed in brief.

6.7.2 Lasers in Microelectronics

In electronics industry lasers are widely used in processing microelectronic components and integrated circuits. Few applications of laser in the field of microelectronics are listed below.

1. It is used for trimming of components of the IC's with higher accuracy.
2. Lasers are used in cutting touch pads for cell phones.
3. Using laser, battery cathodes can be cut from aluminum foil and also to cut dielectric from paper for use in cardiac pacemaker implants.
4. Other applications of lasers in the microelectronics field include; high speed printing (laser printers), laser pointers (an eye catching pointing device used in presentations and visual demonstration), all thin film circuit fabrication, as a source in LAN's (Local Area Network), laser evaporation, bar code scanners etc.

CHARACTERISTICS OF LASERS

5.11

The main characteristics of laser radiations are (a) monochromaticity, (b) coherence, (c) high directionality, and (d) extreme brightness.

(a) **Monochromaticity** The laser light is almost perfectly monochromatic, much more monochromatic than that of any conventional monochromatic source. The optical resonator formed by two mirrors sustains only specific frequencies from the broad band. The mirrors convert the continuous emission band

into a group of sharply defined discrete lines. By some suitable frequency-selective mechanism, a single line can be selected out from these lines.

(b) **Coherence** The beam of laser light is completely coherent, i.e., all waves are exactly in phase with one another. An interference pattern can be obtained also using beams from separate lasers. This is inherent in the process of stimulated emission which is responsible for laser action.

(c) **High Directionality** A laser emits light only in one direction. The width of a laser beam is extremely narrow and hence a laser beam can travel to long distances without spreading. They can be brought to an extremely sharp focus.

(d) **Extreme Brightness** Extreme brightness is the unique property of a laser beam. Due to high energy density and directional property, it can vaporise even the hardest metal; a laser beam can produce temperatures of the order of 10^4 °C at a focused point. Further, a one milliwatt radiation of a helium–neon laser appears brighter than the sun.

USES OF LASERS

5.12

The most significant applications of lasers include telecommunications, medicine, manufacturing, the military, consumer goods and basic research.

(1) **Telecommunications** Telecommunications is one of the most important uses of lasers. Light from diode lasers is modulated by the signal that contains the information, which is then transmitted through a network of optical fibres. Future developments will most likely include tuneable lasers compatible with wavelength division multiplexing, a technique which will even further enlarge the bandwidth of fibre optic systems.

(2) **Medicine** Lasers are frequently used in surgery and medical diagnostics. Surgeons can now use (carbon dioxide) lasers as bloodless scalpels because the optical beam cauterises the incision as it is made. Lasers are also used to open up clogged arteries using a technique called **laser angioplasty**. Cancer treatment is a small but promising field that, today, is based on photodynamic therapy. In terms of the number of procedures, the most important growth areas are eyesight correction and cosmetic surgery. Cosmetic surgery also relies heavily on lasers for removal of pigments, tattoos, unwanted hair, spider veins, and skin resurfacing.

(3) **Manufacturing** The automotive industry relies greatly on high power CO_2 lasers for cutting and welding metals. Nd:YAG lasers are used in drilling holes, spot welding and marking. The printing industry uses a range of lasers, mostly ion lasers, but the trend is to use high-power diode lasers. The semiconductor industry is a major user of ultraviolet (excimer) lasers for photolithography. Photolithography plays an important role in the manufacture of integrated circuits with hundreds of millions of transistors on a semiconductor chip. The entertainment industry (i.e., laser light shows) is based on ion lasers.

(4) **Military** So far, (mercifully) lasers have been found to make poor weapons. On the other hand, they are extensively used in guiding missiles to their destination, range finders and other target designators. It is suggested that in future 'star-wars' applications lasers will be used in space-based weapons and other airborne systems.

(5) **Consumer Goods** DVDs, CD-ROMs, optical discs, etc, all rely on diode lasers. Big changes are coming with the development of laser diodes emitting in the blue and ultraviolet end of the spectrum. Shorter wavelengths (eg., blue and UV) will enable optical storage devices to pack more information per disc. Diode lasers are also at the heart of laser printers, barcode scanners, laser pointers, security devices, etc.